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Research article

Nature-based solutions for natural hazards in Costa Rica: Evidence, gaps, and challenges

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ABSTRACT

Nature-based Solutions (NbS) are increasingly promoted to reduce disaster impacts while providing biodiversity, climate, and social co-benefits, yet evidence in tropical developing contexts remains fragmented. This study presents a systematic literature review of NbS for natural-hazard management in Costa Rica, combining a bibliometric mapping of peer-reviewed research with a qualitative synthesis of national experiences reported in scientific and technical sources. Searches in Scopus, Web of Science, and Google Scholar were complemented by targeted retrieval from Costa Rican university repositories and institutional platforms. Eligible documents explicitly describe NbS or ecosystem-based approaches linked to hazard mitigation, adaptation, or disaster risk reduction and provided sufficient information to classify hazard focus, ecosystem context, and intervention type. Evidence was coded using internationally recognized NbS frameworks and grouped by hazard category, ecosystem/landscape setting, NbS type, reported indicators, benefits, and implementation barriers. Results indicate a marked increase in NbS-related publications and project reporting since 2007. Documented interventions are concentrated in watershed and forested landscapes and mainly target flood regulation, erosion control, and slope stabilization through restoration, riparian buffers, reforestation, and agroecological or green infrastructure measures. Outcomes are commonly reported using hydrological and ecological indicators, while standardized evaluation of effectiveness, equity, and long-term maintenance is less consistent. Key constraints include fragmented governance, limited financing for monitoring and upkeep, and uneven integration of NbS into territorial planning. The review identifies priorities to strengthen NbS mainstreaming in Costa Rica, including improved evaluation protocols, cross-sector coordination, and clearer incorporation of co-benefits into planning and investment decisions.

1. Introduction

Natural hazards including floods, landslides, droughts, storms, and coastal erosion cause significant and growing damage to human societies, economies, and ecosystems across the globe. These hazards increasingly interact with climate change to amplify risks, creating cascading effects that threaten critical infrastructure, disrupt livelihoods, displace populations, and undermine biodiversity and ecosystem services [20]. Global assessments, such as the IPCC [47] and GAR [68] reports, have emphasized how climate-exacerbated disasters are reversing development gains, particularly in the Global South [49], while straining the capacity of humanitarian systems and national governments to respond effectively [40].

Developing and tropical countries often suffer disproportionately from these impacts due to a combination of geographic exposure, structural inequalities, and limited adaptive capacity (Quesada-Román, 2026). These regions tend to host high concentrations of vulnerable populations living in informal settlements, often located in floodplains, unstable hillsides, or drought-prone areas. Inadequate infrastructure, weak institutional planning, and limited access to early warning systems further compound risk [19]. Moreover, economic reliance on climate-sensitive sectors such as agriculture, fisheries, and tourism exacerbates exposure and heightens the sensitivity of livelihoods to environmental shocks [38].

In this context, Nature-based Solutions (NbS) have emerged as a promising paradigm for addressing the complex, interlinked challenges of disaster risk, climate change, and biodiversity loss [55]. Defined broadly as actions that protect, sustainably manage, and restore natural

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or modified ecosystems to address societal challenges, NbS offer a suite of benefits beyond hazard mitigation, including water security, carbon sequestration, biodiversity enhancement, and improvements in physical and mental well-being [70]. They encompass a range of strategies, such as wetland restoration, mangrove conservation, slope revegetation, urban green infrastructure, agroecological farming, and watershed rehabilitation [69]. These interventions reduce hazard exposure and vulnerability by restoring the regulating functions of ecosystems attenuating flood peaks, reducing erosion, stabilizing slopes, and enhancing groundwater recharge while simultaneously delivering long-term ecological and social co-benefits [7,70].

Compared to traditional "gray" infrastructure, NbS are often more cost-effective, flexible, and resilient under future climate scenarios. Their capacity to adapt and evolve with environmental change makes them particularly attractive for climate adaptation and disaster risk reduction (DRR) frameworks [70]. However, despite growing international support from the Paris Agreement, the Sendai Framework, and the Kunming-Montreal Global Biodiversity Framework implementation of NbS at meaningful scales remains uneven, especially in countries with limited financial, technical, or institutional resources [40].

Costa Rica represents a compelling case for exploring the role of NbS in risk reduction. The country is characterized by high biological and ecosystem diversity, ranging from tropical rainforests and mangrove estuaries to mountain cloud forests and dry Pacific woodlands. It is also acutely exposed to multiple natural hazards due to its position at the intersection of tectonic plates, tropical storm paths, and steep, erosion-prone landscapes. While Costa Rica has earned international recognition for its environmental leadership through pioneering policies in forest conservation, protected areas, and payments for ecosystem services, it continues to face recurrent challenges related to flooding, landslides, droughts, and coastal degradation ([12,50]; Acosta-Quesada and [51, 55]).

This paper hypothesizes that Costa Rica's socio-ecological conditions, institutional strengths, and ongoing exposure to natural hazards make it a valuable context to examine how NbS can be implemented, scaled, and integrated into national risk reduction and adaptation strategies. Specifically, the study aims to systematically identify, classify, and analyze NbS that have been applied or proposed for Costa Rica. Drawing from international frameworks, peer-reviewed literature, policy documents, and local case studies, the paper explores both the effectiveness of NbS and the barriers and opportunities for their mainstreaming.

The findings are intended to inform policymakers, practitioners, and researchers about how NbS contribute to sustainable risk reduction in Costa Rica and beyond. More broadly, the insights will support other developing and tropical countries in adopting NbS as strategic tools that bridge climate adaptation, ecosystem management, and sustainable development objectives under increasingly uncertain environmental futures.

2. Materials and Methods

2.1. Study design and scope

This study was designed as a systematic literature review with an exploratory–descriptive orientation to identify, classify, and synthesize NbS used or proposed for the management of natural hazards in Costa Rica. A systematic literature review was selected because research on Nature-Based Solutions in Costa Rica is dispersed across multiple disciplines, publication types, and institutional sources, making it difficult to obtain a comprehensive understanding through conventional narrative reviews. This approach provides a transparent, reproducible, and structured framework for identifying, screening, and synthesizing heterogeneous evidence while minimizing selection bias. Furthermore, because the objective of this study was not to evaluate the effectiveness of a single intervention but rather to characterize the diversity, distribution, and implementation trends of NbS across different hazards and ecosystems, a systematic review represents the most appropriate methodological approach for generating an integrated national assessment and identifying research and policy gaps. The review was structured to capture both scientific evidence and implementation-oriented experiences by integrating peer-reviewed publications with policy and technical documentation. The scope included NbS addressing hazards relevant to the country such as floods, landslides, drought and water scarcity, coastal erosion, soil erosion, and urban climatic hazards while allowing the inclusion of closely related risk drivers frequently discussed in NbS studies, including climate variability/change and ecosystem degradation [21,40,60,62,63,70]

2.2. Search strategy and information sources

A systematic search was conducted across three complementary evidence streams. First, international peer-reviewed literature was retrieved from Scopus and Web of Science to ensure coverage of indexed research and globally comparable evidence. Second, Google Scholar was used to broaden retrieval to regional journals and applied outputs that are often underrepresented in indexing databases, including theses and institutional reports. Third, targeted searches were performed in Costa Rican university repositories and relevant institutional platforms, including government agencies and programs e.g., MINAE/SINAC (Ministry of Environment/Natural Conservation Areas Service), CNE (National Commission of Emergencies and Prevention), AyA (National Institute of Drainage and Sewage), IMN (National Meteorology Institute), INVU (National Institute of Housing and Urbanism) as well as NGO/cooperation platforms where NbS projects are commonly documented. Search strings combined NbS-related terms (e.g., "nature-based solutions", "ecosystem-based approaches", "ecosystem-based adaptation", "green infrastructure"), restoration-related terms (e.g. adaptation, restoration) with hazard-specific keywords and the geographic identifier "Costa Rica," using Boolean operators to refine retrieval.

2.3. Eligibility criteria

Eligibility criteria were established as a priori to ensure consistent selection, following Reid et al. [60]. Documents were included when they explicitly addressed NbS or ecosystem-based approaches and presented a stated or clearly implied function related to hazard mitigation, adaptation, disaster risk reduction. To ensure geographic relevance, the primary focus needed to be Costa Rica; however, selected studies from comparable tropical contexts were retained only when used to frame transferable lessons for Costa Rican practice and were explicitly flagged as non-Costa Rican evidence. Records also needed to provide sufficient detail to extract at minimum the NbS intervention type, the ecosystem or territorial context, and the hazard addressed. Peer-reviewed articles, theses, technical reports, policy or planning instruments, and project evaluations were all considered eligible, reflecting the applied nature of NbS implementation. Duplicates, purely conceptual documents without identifiable interventions, documents unrelated to hazard management, and texts lacking sufficient reporting detail for coding were excluded.

2.4. Data extraction, coding, and classification framework

Screening followed a transparent multi-stage procedure consistent with systematic review practice. After consolidating records from all sources (69 in total) and removing duplicates, titles and abstracts (or executive summaries in technical reports) were screened against the eligibility criteria. Records that passed the initial screening were then evaluated through full-text assessment to confirm relevant information. Exclusions at the full-text stage were justified and documented to maintain transparency.

A standardized extraction and coding template was developed to ensure that evidence could be synthesized consistently across

heterogeneous document types. For each document, bibliographic information (e.g., year and document type) was extracted, and thematic variables were coded to match our analytical structure (described next). Specifically, each document was classified by year and according to the primary hazard addressed, the ecosystem or landscape context, the NbS intervention type, the categories of indicators reported for monitoring or evaluation, the types of benefits described, and the limitations or barriers to implementation and scaling. NbS classification was guided by internationally recognized standards, including the IUCN Global Standard for NbS [33], complemented by ecosystem-based disaster risk reduction typologies used in policy and applied research.

2.5. Synthesis approach and analytical outputs

Evidence was synthesized using a combined descriptive and narrative approach because the retrieved studies differed substantially in objectives, methodologies, indicators, ecosystems, and hazard types, precluding quantitative meta-analysis while allowing robust thematic synthesis of the available evidence (Fig. 1; [46]). Quantitative summaries were produced as frequency distributions (counts and percentages) to characterize patterns by publication year and across the thematic variables described above. In parallel, a narrative synthesis was developed to interpret the observed patterns, highlight gaps and underrepresented hazards or ecosystems, and connect the evidence base to implementation realities and policy relevance in Costa Rica. Where quantitative performance outcomes were reported (e.g., runoff regulation, sediment retention, slope stabilization proxies, water storage, or urban cooling), these were extracted and recorded; however, due to heterogeneity in study designs and metrics, findings were synthesized qualitatively rather than through meta-analysis [22,44,64].

Visual materials representing most common NbS for Costa Rica identified in the bibliometric analysis were produced using AI-assisted tools (ChatGPT) and subsequently refined and edited by a graphic designer. Given the diversity of evidence sources and variation in reporting detail, particularly in grey literature, this review emphasizes evidence coverage, thematic trends, and implementation insights rather than effect-size estimation. Potential publication and reporting bias were reduced by triangulating peer-reviewed studies with institutional and project documentation and by documenting exclusions and their reasons.

3. Results and Discussion

3.1. Years

Overall, we found a clear increase in the number of publications between 2007 and 2025, with highest numbers particularly after 2019. Although this trend suggests growing interest and activity in Nature-Based Solutions, it should not be interpreted solely as evidence of increased implementation. Rather, the observed pattern likely reflects a combination of factors, including a genuine expansion of NbS initiatives, increased research activity, improved documentation by government agencies and NGOs, greater availability of grey literature, and enhanced accessibility of scientific publications through digital repositories. The observed increase in publications after 2015 corresponds with major climate and NbS policy milestones. The Paris Agreement spurred international funding streams, which we hypothesize encouraged the publication of adaptation-oriented studies in that period [40]. The temporal distribution of NbS publications in Costa Rica (Fig. 2) reveals a notable increase in research output after 2015, with peaks in 2021 (n = 8), 2022 (n = 7), and 2025 (n = 8). This trend aligns with growing national and international interest in integrating ecosystem-based approaches into climate adaptation and disaster risk reduction frameworks [40,70].

In addition, there has been an increasing trend in the quantity of intense hurricanes in the Caribbean Sea since the 1970s, which intensifies with La Niña conditions, leading to landslides and floods in the country [31,43,52,54]. Consequently, the period following Hurricane Otto and Tropical Storm Nate marked a policy shift toward integrating ecological buffers into risk management [39]. The sustained publication activity after 2020 probably reflects both increased implementation of nature-based interventions and greater scientific attention devoted to documenting and evaluating these initiatives. As NbS became increasingly incorporated into climate adaptation, ecosystem restoration, and disaster risk reduction policies, more projects were implemented and simultaneously accompanied by technical reports, monitoring efforts, and peer-reviewed publications. Consequently, publication trends should be interpreted as an indicator of growing visibility and documentation of NbS rather than as a direct measure of implementation alone [58,6]. The publication peaks in 2021–2025 may also reflect increased investment in urban green infrastructure (Corrales et al., 2019; [6]), coastal adaptation [12], and NbS policy mainstreaming [70]. The slight decline observed in 2024 likely reflects normal year-to-year variation in publication output rather than a reduction in NbS implementation, particularly considering the renewed increase in 2025.

3.2. Hazard type

In close agreement with Quesada-Román et al. [53], environmental degradation and ecosystem loss emerge as the most frequently addressed hazard category in NbS applications in Costa Rica (n = 24; ~35%), encompassing processes such as riverine and coastal pollution. This is followed by climate change and climate variability (n = 19; ~28%), including sudden floods, tropical cyclones, and droughts, while hydrometeorological hazards rank third (n = 14; ~23%), a pattern that has remained stable over the past decade [9]. In contrast, governance-related challenges (n = 7; ~10%) and geomorphological hazards (n = 5; ~7%) are comparatively underrepresented in the literature, despite their relevance to riverine and wetland dynamics, as well as soil and coastal erosion processes (Fig. 3).

The predominance of climate-related and ecosystem degradation

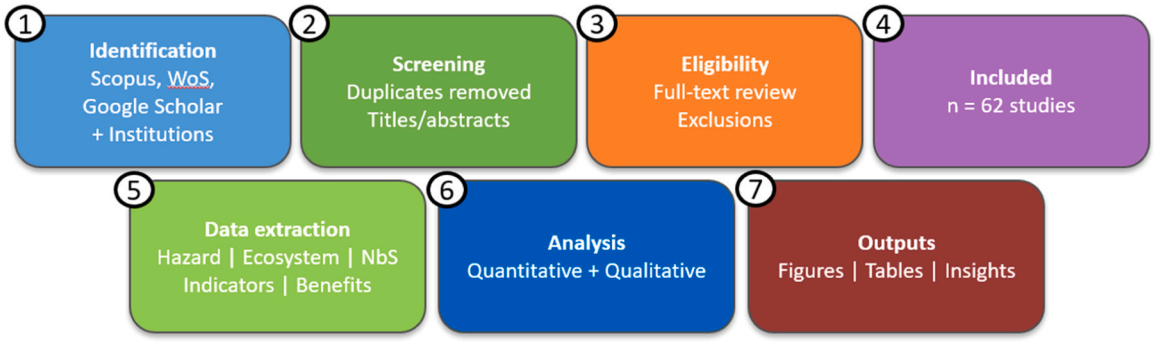


Fig. 1. PRISMA-enhanced framework used to identify, screen, and classify Nature-based Solutions (NbS) for natural hazards in Costa Rica [46].

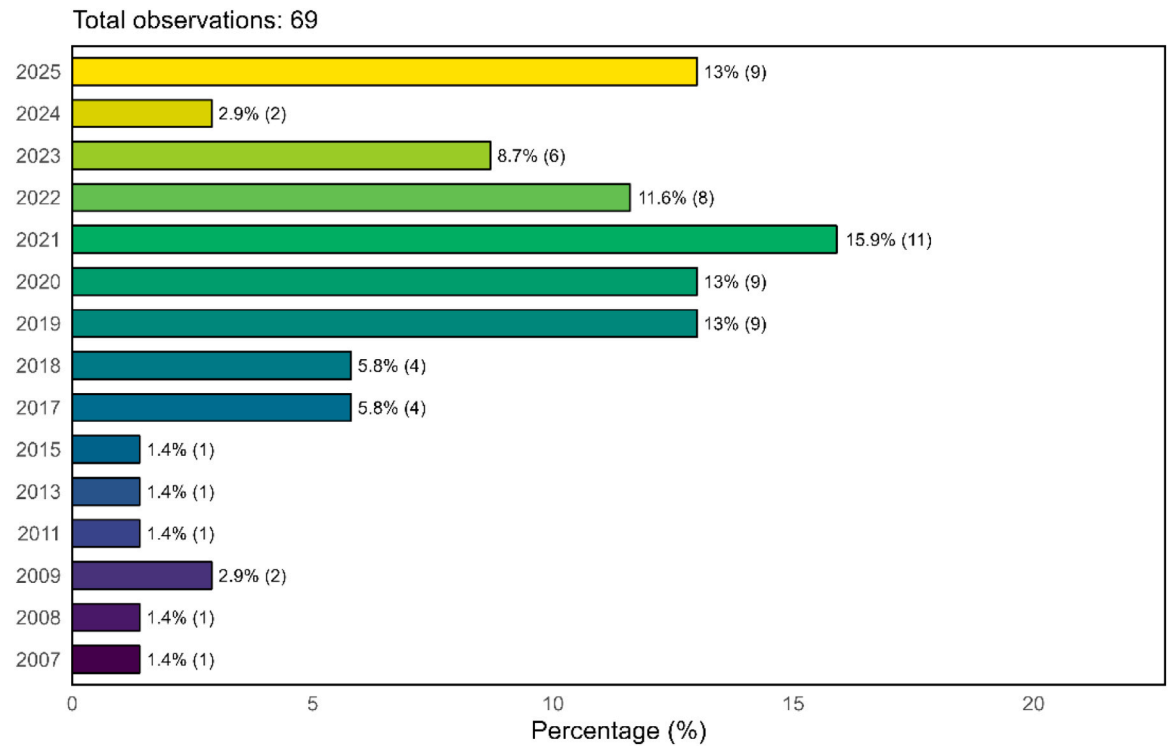


Fig. 2. Annual scientific and grey literature production on Nature-based Solutions (NbS) for natural hazards in Costa Rica (2007–2025).

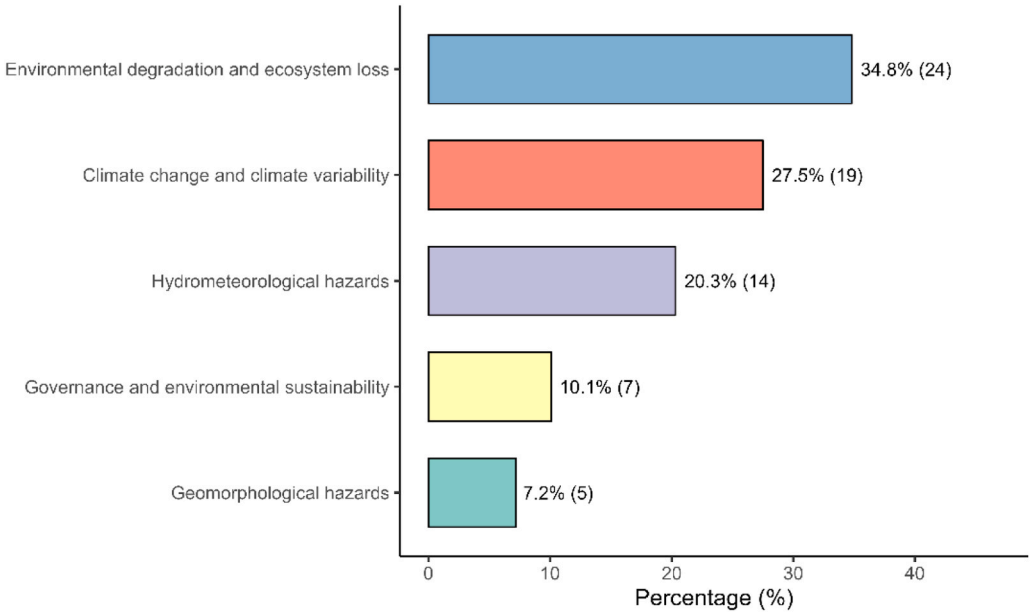


Fig. 3. Most common natural hazards addressed by Nature-based Solutions (NbS) in Costa Rica, as reported in the literature published between 2007 and 2025.

hazards reflects Costa Rica's exposure to variability in rainfall regimes and temperature extremes, with cascading impacts on ecosystems and livelihoods [57]. These dynamics are likely reinforced by land-use change, including the decline of primary forests, which contributes to both global climate forcing and local hazard intensification. At the same time, localized stewardship systems provide evidence of resilience pathways: in the Talamanca Indigenous Territory, Bribri and Cabécar management practices illustrate how long-standing socio-ecological approaches can sustain biodiversity and buffer environmental change [19,20].

3.3. Ecosystem type

Urban and interurban ecosystems dominate the NbS literature in Costa Rica (n = 27; 31.9%), reflecting both the concentration of population and research capacity in the Greater Metropolitan Area and the prominence of municipal initiatives promoting green infrastructure and urban biodiversity [17,48]. Rural and agricultural systems follow (n = 14; 20.3%), alongside mixed or institutional ecosystems (n = 9; 13%), while forests and terrestrial natural areas (n = 8; 11.6%) continue to feature due to the country's longstanding emphasis on landscape restoration, biological conservation, and corridor management [32,66].

Rivers and waterways (n = 7; 10.1%) and coastal or marine systems (n = 4; 5.8%) are comparatively underrepresented, despite their ecological and socio-economic importance (Fig. 4).

This distribution underscores a strong urban bias in NbS research and implementation, while also revealing gaps in freshwater and coastal ecosystem coverage [12,42]. At the same time, the representation of forest ecosystems reflects their continued centrality in national conservation policy and practice. Within these systems, riparian forests are frequently highlighted as sensitive indicators of biodiversity responses to bioclimatic variability, particularly for wild mammal assemblages [23,36]. Overall, the observed pattern suggests that the spatial distribution of NbS evidence aligns with policy and research priorities that remain disproportionately urban-centered, pointing to structural imbalances in conservation investment and knowledge production across ecosystem types.

Furthermore, coastal and riverine ecosystems are considerably vulnerable to flooding, erosion, sea-level rise, and pollution [2]. NbS such as mitigation against coastal erosion from sandy beaches of South Caribbean using mangrove or community engagement are punctual examples [5]. Several studies have shown that coastal erosion along the coastline has impacted local towns including vegetation, housing infrastructure and possible pollution of groundwater. Moreover, coastal landscape, reef and mangrove biodiversity changes have promoted actions orienting nature base solutions towards addressing coastal erosion [10,12].

3.4. Nature-based solution type

Ecological restoration is the most common NbS type in the country (n = 18, 26.1%), emphasizing the importance of rehabilitating degraded ecosystems for hazard mitigation and biodiversity recovery. Policy and Ecosystem Services approaches rank second (n = 17, 24.6%), followed by Unclassified cases (n = 13, 18.8%), then by Green Infrastructure (n = 9, 13%), and Agroecological practices (n = 6, 8.7%), Ecosystem-based Adaptation (n = 5, 7.2%), and Coastal Solutions (n = 1, 1.4%) are less represented (Fig. 5). This distribution reflects Costa Rica's strong institutional frameworks for ecosystem services and restoration [4,64] and the underlying assumption that these arrangements trickle down into field implementation. However, the effectiveness of Nature-based Solutions (NbS) in Costa Rica is highly context-dependent and varies according to ecosystem characteristics, geographical setting, and implementation scale. For example, mangrove restoration and conservation have proven particularly relevant in coastal areas such as the Gulf of Nicoya and Térraba-Sierpe, where they contribute to shoreline stabilization, flood mitigation, carbon sequestration, and biodiversity conservation. In contrast, forest restoration and agroforestry systems have been more widely implemented in mountainous watersheds of the Central Volcanic Range and Talamanca region, where they support erosion control, slope stabilization, water regulation, and ecosystem connectivity (REFS). These examples highlight that NbS must be tailored to local environmental and socio-economic conditions to maximize their effectiveness.

Despite approximately 92% of Costa Rica's national territory being marine, terrestrial protected areas historically received greater attention and investment due to the country's conservation model, which was initially developed around forest protection, national parks, and watershed conservation. As a result, terrestrial protection currently exceeds 25% of the land area, whereas marine protected areas, although expanding rapidly in recent years, have traditionally covered a smaller proportion of the Exclusive Economic Zone (REFS). This pattern reflects historical conservation priorities rather than the relative importance of marine ecosystems, which are increasingly recognized as critical for biodiversity conservation, climate adaptation, fisheries sustainability, and coastal resilience.

The strong emphasis on ecological restoration corresponds to Costa Rica's longstanding investment in forest recovery, reforestation, and ecological rehabilitation [4,59]. While this reflects national conservation strengths, it may narrow the operational diversity of NbS approaches in practice (Fig. 5). Green infrastructure remains less prevalent in the literature despite rapid urbanization and its clear need for improved stormwater management and thermal regulation [29,3,44]. Coastal NbS appear critically underrepresented, which contrasts with global recommendations identifying mangrove, dune, and coral restoration as essential adaptation measures for tropical coastal nations [12, 67]. This is a significant strategic gap and reflects the national lack of importance given to marine ecosystems and their biodiversity and richness.

In Costa Rica, the most mentioned and implemented bS reflect both the country's wide range of hydrometeorological and geomorphological hazards and its strong environmental management tradition (Fig. 6). Among the most frequent measures are bioengineering approaches, such as the use of vegetated terraces and slope stabilization techniques to

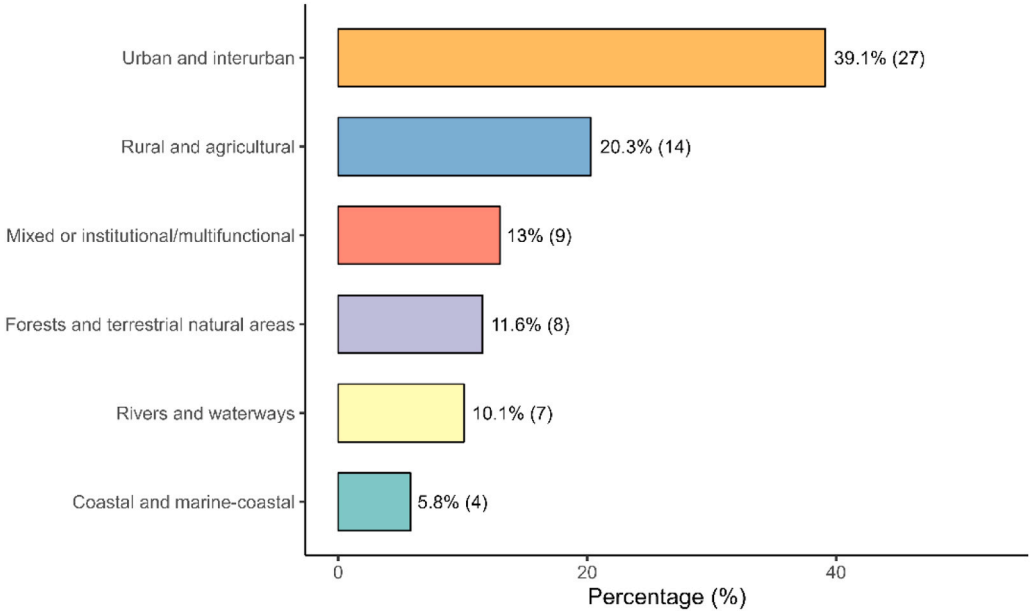


Fig. 4. Most common ecosystem types where Nature-based Solutions (NbS) have been documented in Costa Rica.

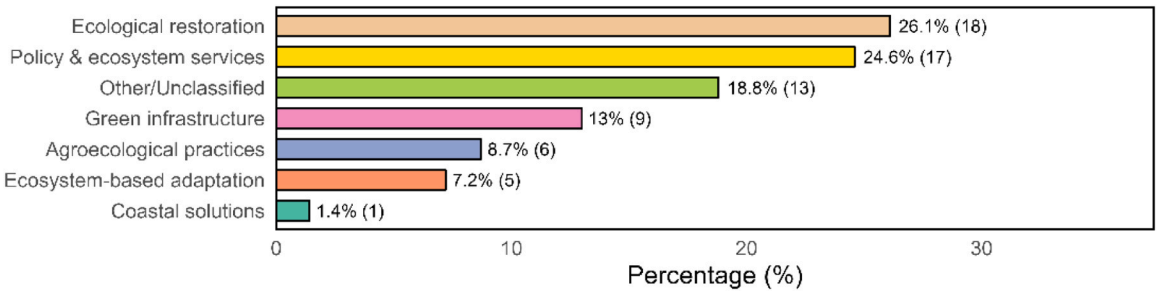


Fig. 5. Frequency of Nature-based Solution (NbS) types documented in Costa Rica between 2007 and 2025.

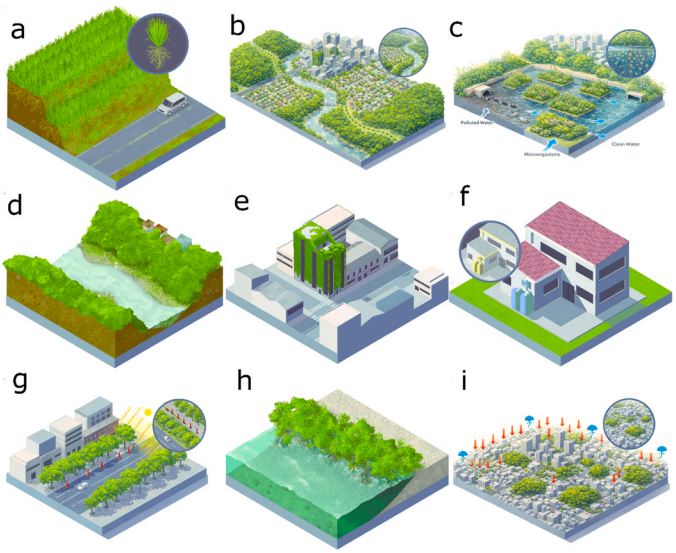


Fig. 6. Representative examples of the most common Nature-based Solutions (NbS) identified in Costa Rica: a) bioengineering using terraces to reduce landslides, b) urban biological corridors, c) bioremediation lagoons, d) riparian linear corridors, e) green infrastructure, f) rainwater storage, g) street trees to mitigate high urban temperatures, h) mangrove reforestation to reduce coastal erosion, i) urban forests to reduce heat waves.

reduce landslide occurrence [50], particularly in steep and highly modified mountainous regions [1]. Urban biological corridors and riparian linear corridors along rivers also emerge as key strategies to enhance ecological connectivity, regulate surface runoff, and reduce exposure to flooding [8,24,37].

Other widely reported NbS include green infrastructure and rainwater storage systems, which contribute to urban adaptation while addressing risks linked to extreme rainfall events. In cities, street trees and the expansion of urban forests are increasingly recognized as essential tools to mitigate high temperatures and heat waves, providing direct benefits for public health and well-being [17]. In coastal areas, mangrove reforestation stands out as one of the most important solutions to reduce erosion and buffer the impacts of storm surges and coastal flooding. Finally, bioremediation lagoons represent a relevant NbS for natural water treatment and environmental quality improvement, integrating risk reduction with ecosystem services in peri-urban and rural contexts [39].

3.5. Indicator type

NbS monitoring frameworks in Costa Rica are strongly dominated by ecological and biodiversity indicators (n = 34; 49.3%), reflecting the central role of ecological and forestry institutions such as SINAC, CATIE, and UCR in knowledge production and implementation (Fig. 7). This is followed by indicators related to water resources (n = 12; 17.4%) and ecosystem services or sustainability metrics (n = 10; 14.5%). In contrast, social, institutional, and governance indicators (n = 7; 10.1%),

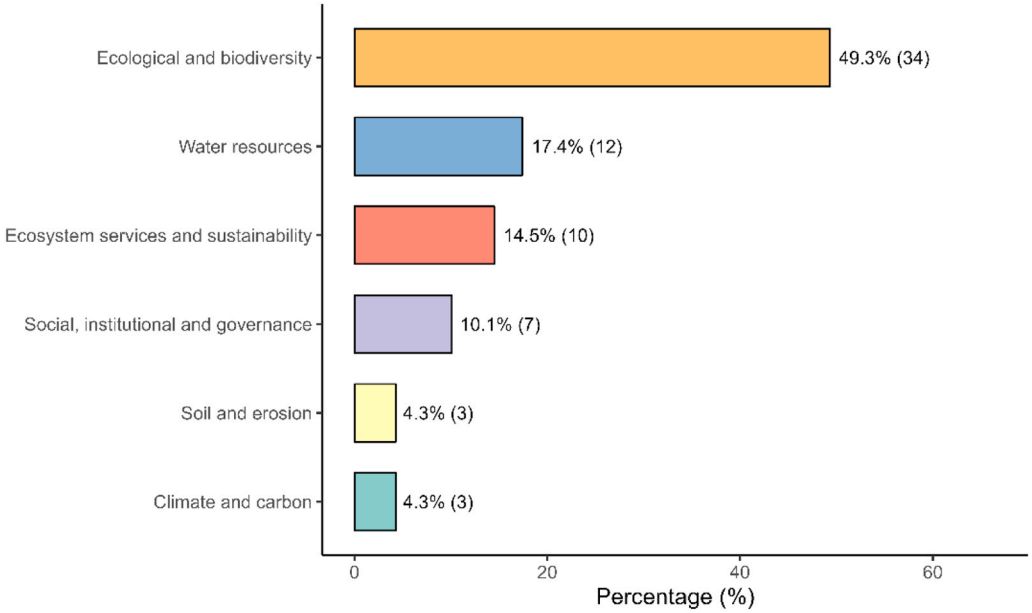


Fig. 7. Types of indicators used to evaluate the effectiveness of Nature-based Solutions (NbS) in Costa Rica.

as well as soil and erosion indicators (n = 3; 4.3%) and climate or carbon indicators (n = 3; 4.3%), are comparatively underrepresented [16,30,41,44]. In general terms, literature does not evidence the monitoring system types, that is the main uncertainty of those implementations in the country.

This distribution indicates a predominantly biophysical orientation in NbS monitoring, which, while ensuring robust ecological assessment, constrains the evaluation of socio-economic outcomes and governance performance that are critical to long-term effectiveness. Notably, the limited use of soil and erosion indicators diverges from the well-established importance of erosion control and slope stabilization in Costa Rica's hazard-prone landscapes [13,35,62]. Expanding the integration of soil-related metrics would enhance the capacity of NbS monitoring systems to capture risk reduction outcomes and support sustainable land management practices, particularly in agricultural and mountainous regions [56].

These indicators reflect an overall conservation and ecological scope, focusing on biodiversity and the richness of Costa Rican ecosystems. However, ecological indicators alone are insufficient to assess the full effectiveness of Nature-based Solutions (NbS), as their success also depends on how ecosystem functions are integrated into land-use planning, governance processes, and community development strategies. By linking ecological integrity with spatial planning instruments, NbS can simultaneously support biodiversity conservation, ecosystem service provision, and risk reduction while promoting sustainable livelihoods and social well-being. In this context, territorial planning serves as a bridge between ecosystem conservation objectives and societal needs, allowing ecological information to inform land-use decisions and long-term development pathways. Consequently, monitoring frameworks for NbS should combine ecological, territorial, and socio-economic indicators to capture both environmental outcomes and benefits for local communities. Territorial planning oriented toward sustainable development can enhance policies aimed at improving community well-being while respecting ecosystem limits [4].

3.6. Benefit type

Ecological and conservation benefits are the most frequently reported outcomes of NbS in Costa Rica (n = 37; 53.6%), consistent with the predominance of restoration- and forest-based approaches in the literature [14,61]. Water-related benefits follow (n = 15; 21.7%), reflecting the central role of watershed management within national environmental governance frameworks [18,34]. Additional reported benefits include climate regulation (n = 8; 11.6%) and governance or planning outcomes (n = 9; 13.0%), while social well-being (n = 6; 8.7%) and economic benefits (n = 4; 5.8%) are comparatively less emphasized (Fig. 8).

This distribution reinforces the multifunctional character of NbS while highlighting a strong biophysical and environmental bias in reported outcomes. The comparatively limited representation of social and economic benefits suggests gaps in capturing livelihood improvements, equity dimensions, and financial performance, which are critical for sustaining community support and informing policy and investment decisions [15,30,45]. Expanding the integration of socio-economic and governance metrics would strengthen the evidence base for NbS contributions to inclusive development and risk reduction.

In this context, community-based initiatives illustrate the potential to align conservation outcomes with territorial development. Ensuring that benefits are equitably distributed and not disproportionately captured by large-scale commercial interests remains a key condition for long-term socio-ecological sustainability [12,30].

3.7. Limitation type

Spatial and territorial pressures constitute the most frequently cited limitation to NbS implementation in Costa Rica (n = 18; 26.1%), followed by financial and resource constraints (n = 16; 23.2%), technical and implementation challenges (n = 15; 21.7%), institutional and governance barriers (n = 8; 11.6%), social and participation issues (n = 7; 10.1%), and environmental or climatic factors (n = 5; 7.2%) (Fig. 9). This distribution underscores the multi-dimensional nature of constraints affecting NbS scaling, particularly in rapidly urbanizing and resource-constrained contexts [29,65]. It also suggests that NbS implementation is constrained in Costa Rica by institutional factors that shape land use and governance resources allocation, and not by biophysical conditions per se.

The predominance of spatial pressures reflects persistent land scarcity, and competing development demands in urban and peri-urban territories, where land-use conflicts place increasing pressure on ecosystems and local communities null [4,17,44]. Financial constraints further limit implementation at scale, consistent with reliance on external funding mechanisms for NbS deployment [40,6]. In parallel,

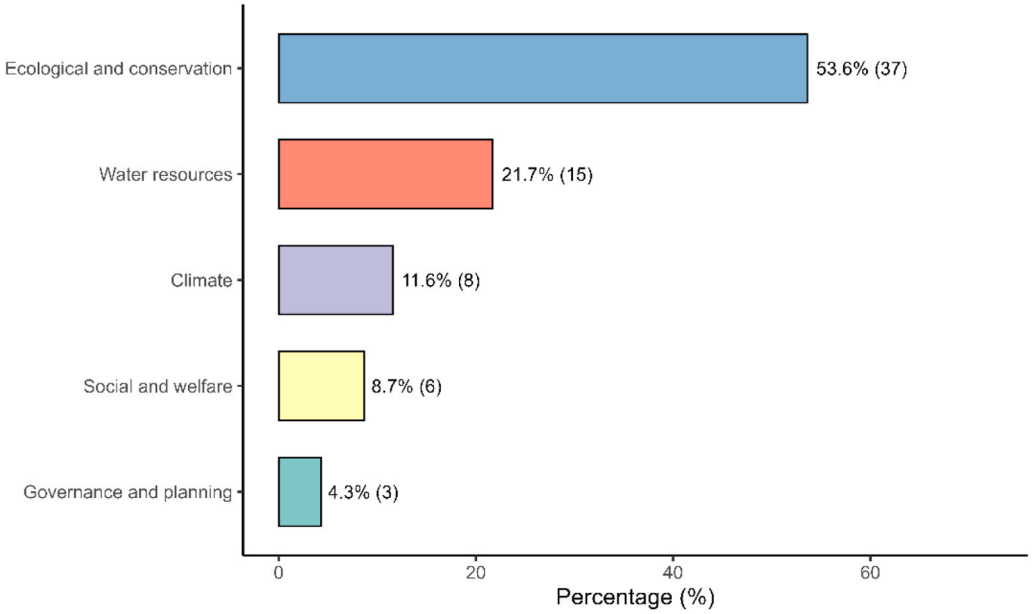


Fig. 8. Reported benefits of Nature-based Solutions (NbS) in Costa Rica across the reviewed literature.

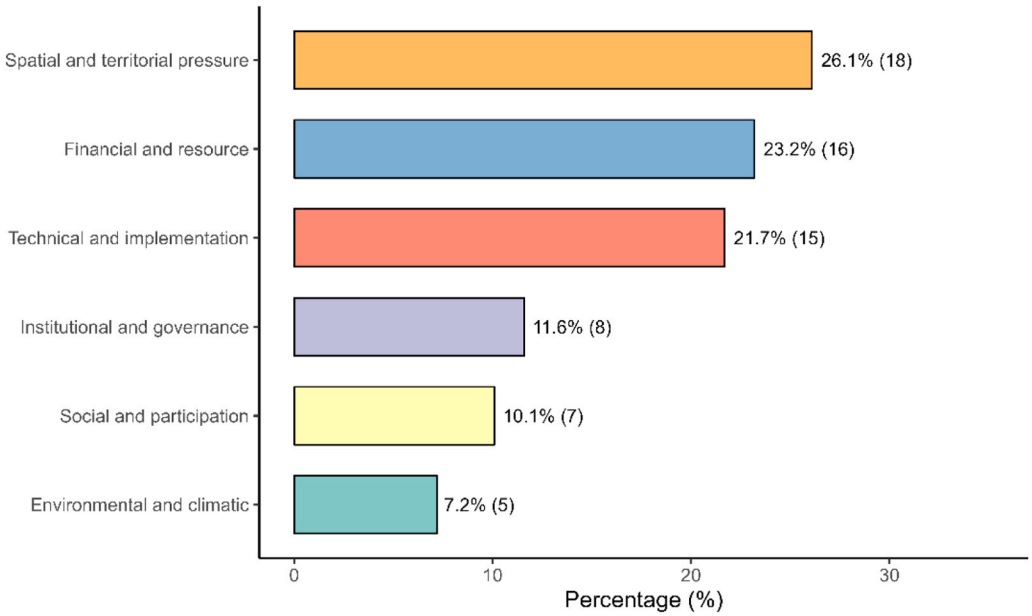


Fig. 9. Main limitations and barriers affecting the implementation of Nature-based Solutions (NbS) in Costa Rica.

technical barriers point to capacity gaps at the municipal level, affecting design, execution, and maintenance of NbS interventions [29,44]. Institutional and governance challenges, including fragmentation among agencies such as MINAE/SINAC (Ministry of Environment/Natural Conservation Areas Service), CNE (National Commission of Emergencies and Prevention), ICE (Costa Rican Electricity Institute), AyA (National Institute of Drainage and Sewage), IMN (National Meteorology Institute), INVU (National Institute of Housing and Urbanism), reinforce coordination bottlenecks, particularly in riverine and coastal management contexts [65]. Although less frequently reported, social participation constraints highlight the need to strengthen community engagement, especially in rural and agroecological systems where local stewardship is critical to long-term effectiveness [11,19].

3.8. Global implications

Although this review focuses on Costa Rica, several findings have broader relevance for countries facing increasing climate-related hazards. Some aspects of the Costa Rican experience are highly context specific. The country's long-standing environmental policies, extensive protected area network, successful forest restoration programs, Payments for Ecosystem Services (PES), and relatively strong environmental institutions have created favorable conditions for implementing Nature-Based Solutions (NbS). These governance characteristics may not yet be present in many low- and middle-income countries, where institutional capacity, financial resources, and long-term environmental planning remain more limited.

Nevertheless, many of the lessons identified in this review are transferable across tropical regions in Latin America, Africa, and Southeast Asia that experience similar socio-ecological conditions. The evidence suggests that ecosystem conservation and restoration can simultaneously reduce floods, landslides, coastal erosion, and drought risks while generating co-benefits for biodiversity conservation, climate change adaptation, water security, carbon sequestration, and local livelihoods. Likewise, integrating NbS into land-use planning, watershed management, coastal zone management, and disaster risk reduction strategies represents an approach that can be adapted regardless of differences in governance structures. Community participation, interdisciplinary collaboration, and the combination of engineering and ecosystem-based approaches also emerge as broadly applicable principles for successful NbS implementation.

Looking ahead, the relevance of NbS is likely to increase as climate change intensifies hydro-meteorological extremes across tropical developing countries. Advances in climate prediction, including artificial intelligence and deep learning techniques, are improving the capacity to project future climatic conditions and extreme events at regional and local scales [26,25,27,28]. These developments can support more effective identification of priority areas for ecosystem restoration, watershed management, and climate adaptation, thereby strengthening evidence-based NbS planning. Consequently, the Costa Rican experience provides not only a national assessment but also a useful framework for countries seeking to mainstream NbS into disaster risk reduction and climate adaptation policies under increasingly uncertain future climate scenarios.

4. Conclusions

Nature-based Solutions are rapidly emerging as key strategies for managing natural hazards in Costa Rica, reflecting the country's rich ecological diversity and long-standing environmental governance frameworks. This review demonstrates that documented NbS interventions are primarily concentrated in watershed and urban ecosystems, targeting flood regulation, erosion control, slope stabilization, and climate-related urban hazards through restoration, riparian buffers, reforestation, and green infrastructure. While ecological benefits and biodiversity outcomes are widely reported, the evidence base remains uneven in terms of standardized monitoring, long-term effectiveness assessment, and integration of social equity and governance dimensions. Overall, Costa Rica provides an important example of NbS potential in tropical developing contexts but also highlights persistent gaps that limit broader implementation and scaling.

Future efforts should prioritize the development of robust evaluation protocols that integrate biophysical, socioeconomic, and governance indicators, aligned with international NbS standards and disaster risk reduction frameworks. Strengthening cross-sector coordination among environmental agencies, municipalities, and risk management institutions will be essential to mainstream NbS into territorial planning and infrastructure investment. In addition, coastal and freshwater NbS require greater research and implementation attention, given Costa Rica's exposure to erosion, sea-level rise, and hydrometeorological extremes. Expanding long-term financing mechanisms, participatory governance, and equity-focused approaches will be critical for ensuring

that NbS deliver sustainable risk reduction benefits while supporting resilience, biodiversity conservation, and inclusive development.

Although some of these findings are shaped by Costa Rica's unique institutional context, including its long-standing conservation policies and environmental governance, many of the underlying principles are transferable to other tropical and temperate developing countries. Integrating ecosystem conservation and restoration into disaster risk reduction, climate adaptation, and territorial planning, supported by interdisciplinary collaboration, evidence-based decision-making, and stakeholder participation, represents a broadly applicable pathway for mainstreaming NbS. As climate change intensifies hydro-meteorological hazards across the tropics, the lessons synthesized in this review provide a useful framework for guiding future research, policy development, and implementation efforts beyond Costa Rica.

CRediT authorship contribution statement

Adolfo Quesada-Román: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Miguel Cifuentes-Jara:** Writing – review & editing, Writing – original draft, Validation, Investigation. **Manuel Peralta-Reyes:** Writing – original draft, Resources, Investigation, Formal analysis.

Declaration of Competing Interest

We approve the final version of the submitted manuscript. We warrant that the article is the authors' original work, has not received prior publication and is not under consideration for publication elsewhere.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nxsust.2026.100417.

References

[1] M. Acosta-Quesada, A. Quesada-Román, Landslides and flood hazard mapping using geomorphological methods in Santa Ana, Costa Rica, Int. J. Disaster Risk Reduct. 113 (2024) 104882.
[2] J.F. Acuña-Piedra, A. Quesada-Román, Multidecadal biogeomorphic dynamics of a deltaic mangrove forest in Costa Rica, Ocean. & Coast. Manag. 211 (2021) 105770.
[3] G.E. Alvarado, A. Quesada-Román, Anthropogenic Geomorphology of Costa Rica. Landscapes and Landforms of Costa Rica, Springer International Publishing, Cham, 2024, pp. 259–279.
[4] D.F. Avendaño-Leadem, B.C. Cedeño-Montoya, M.S. Arroyo-Zeledón, Integración del concepto de servicios ecosistémicos en la planificación territorial, Rev. Geogr. áfica De. Am. érica Cent. (65) (2020) 63–90.
[5] G. Barrantes-Castillo, J.F.V. Calderón, F.D. Barros, E.B. Alvarado, C.A. Quesada, Análisis espacio temporal de la erosión y acreción costera en Gandoca, Limón, Costa Rica (2005-2019) e implicaciones socioambientales desde la perspectivacomunitaria, Rev. Cienc. Mar. Y. Coste (2026) 67–87.
[6] D. Borge-Leandro, Infraestructura verde para el control de aguas pluviales como medida de adaptación al cambio climático, Rev. Tecnol. ía En. Marcha 38 (1) (2025) 104–114.
[7] O. Calvo-Solano, A. Quesada-Román, Worldwide research trends and networks on flood early warning systems, GeoHazards 5 (3) (2024) 582–595.
[8] J.E. Calvo-Villalobos, T. Bermúdez-Rojas, H. Vega-Bolaños, Dinámica del uso del suelo y sitios prioritarios para la restauración del Bosque del Corredor Biológico Río Tibás, Costa Rica, Rev. Geogr. áfica De. Am. érica Cent. 62 (2019) 138–163.
[9] D. Campos-Durán, A. Quesada-Román, Impacto de los eventos hidrometeorológicos en Costa Rica, periodo 2000-2015, Geo Uerj 30 (2017) 440–465.
[10] Carvajal Oses, M. (2023). Plan de educación ambiental que apoye la gestión sostenible de los principales servicios ecosistémicos de provisionamiento y culturales en el manglar de Chacarita, Puntarenas. [Tesis de Doctorado]. Tecnológico de Costa Rica, Cartago, Costa Rica. http://hdl.handle.net/11056/26585.
[11] F. Casasola, M. Ibrahim, C. Sepúlveda, N. Ríos, D. Tobar, Implementación de sistemas silvopastoriles y pago por servicios ambientales en Esparza, Costa Rica: Una herramienta de adaptación al cambio climático en fincas ganaderas, Buenas Pr. ácticas Agr. ícolas Para. la Adapt. óN. al Cambio Clim. ático (2009) 9–35.
[12] M. Castillo-Chinchilla, L. Piedra-Castro, J.M. Pereira Chaves, L. Sierra Sierra, Acciones de mitigación y adaptación ante la erosión costera de playas arenosas del Caribe Sur, Costa Rica, Invest. Geogr. áficas (115) (2024), https://doi.org/10.14350/rig.60854.
[13] M. Chinchilla-Ureña, I. Guzmán-Arias, F. Watson-Hernández, C. Vargas-Meneses, Diseño de dos propuestas de obras de conservación de suelos para mitigar erosión en terreno con cárcavas, Zona de Los Santos, Costa Rica, Rev. Tecnol. ía En. Marcha 36 (4) (2023) 124–136.
[14] Cifuentes-Jara, M., Brenes, R., Brenes, C., Corrales, L., Vargas, M., Betbeder, J., Vargas, G., Guerrero, A., & Fung, E. (2021). Los árboles en la ciudad: carbono como servicio ecosistémico urbano. Proyecto Biodiver_City, GIZ. Recuperado de www.atlasverde.org.
[15] Cifuentes-Jara, M., Brenes, R., Brenes, C., Corrales, L., Vargas, M., Betbeder, J., Vargas, G., Guerrero, A., & Fung, E. (2023). Biodiversidad en la ciudad: conectando los espacios verdes. Recuperado de www.atlasverde.org.
[16] Contreras del Valle, M.F., Starnfeld, F. (2022). Guía para la Valoración Económica de Servicios Ecosistémicos Marinos y Costeros. Iniciativa Bio-Bridge, GIZ.
[17] L. Corrales, El papel de la conectividad y la infraestructura verde urbana en la adaptación al cambio climático, Ambientico 272 (2019) 74–83.
[18] A.Y. Espinal Girón, *Evaluación del efecto de soluciones basadas en la naturaleza en el servicio ecosistémico hídrico en la microcuenca del río Torres, San José, Costa Rica.* [Tesis de maestría, Cent. Agron. ómico Trop. De. Invest. óN. Y. Enseñanza (CATIE)] (2022). https://repositorio.catie.ac.cr/handle/11554/2773.
[19] N.L. Espinoza, M. Potma, Respuestas agroecológicas para la resiliencia climática: Dos experiencias en Costa Rica, LEISA Rev. De. Agroecol. ía 36 (2) (2020).
[20] B. Finegan, C. Medellín, D. Delgado, Biodiversidad y servicios ecosistémicos de bosques de montaña de la Cordillera de Talamanca, Costa Rica Invest. óN. Apl. a la gestióN. Para. la Adapt. óN. al Cambio Clim. ático (2020). https://repositorio.catie.ac.cr/handle/11554/10254.
[21] M.D.M. Garro-Quesada, M. Vargas-Leiva, P.O. Girot, A. Quesada-Román, Climate risk analysis using a high-resolution spatial model in Costa Rica, Climate 11 (6) (2023) 127.
[22] S. Geppert, G. Chavarría Solís, Biodiver_City: Fortaleciendo ecosistemas y personas para crear ciudades verdes, Ambientico (280) (2021) 4.
[23] S. Granados-Bolaños, J. Rojas-Jiménez, H. Aguilar-Pérez, A. Quesada-Román, E. Brenes-Mora, A low-cost UAV survey can unravel Baird's Tapir (Tapirus bairdii) trail network dynamics in a neotropical highland forest, Ecol. Inform. 82 (2024) 102764.
[24] S. Granados-Bolaños, N. Surian, C. Birkel, G.E. Alvarado, A. Quesada-Román, J. P. Galve, A. Bonilla-Mata, Fluvial landscapes of Costa Rica: an overview of dynamic rivers, Landsc. Landf. Costa Rica (2024) 91–111.
[25] Q. Guo, Z. He, Z. Wang, Prediction of monthly average and extreme atmospheric temperatures in Zhengzhou based on artificial neural network and deep learning models, Front. For. Glob. Change 6 (2023) 1249300.
[26] Q. Guo, Z. He, Z. Wang, Long-term projection of future climate change over the twenty-first century in the Sahara region in Africa under four Shared Socio-Economic Pathways scenarios, Environ. Sci. Pollut. Res. 30 (9) (2023) 22319–22329.
[27] Q. Guo, Z. He, Z. Wang, Monthly climate prediction using deep convolutional neural network and long short-term memory, Sci. Rep. 14 (1) (2024) 17748.
[28] Q. Guo, Z. He, Z. Wang, S. Qiao, J. Zhu, J. Chen, A performance comparison study on climate prediction in Weifang City using different deep learning models, Water 16 (19) (2024) 2870.
[29] J. Hack, Soluciones basadas en la naturaleza para la restauración fluvial en áreas metropolitanas: Proyecto Urban Water Vision en la cuenca Quebrada Seca–Río Burío, Ambientico 280 (2021) 42–49.
[30] B. Herrera-Fernández, F. Carazo, Sarapiquí resiliente: Hacia un modelo de gestión sostenible de los beneficios de la naturaleza en territorios rurales, Ambientico 268 (2018) 18–25.
[31] H.G. Hidalgo, E.J. Alfaro, A. Quesada-Román, Flood projections for selected Costa Rican main basins using CMIP6 climate models downscaled output in the HBV hydrological model for scenario SSP5-8.5, Hydrol. Res. Lett. 18 (1) (2024) 35–42.
[32] M.S. Isaza, J.J. Campos, C. Prins, R. Villalobos, Evaluación de la restauración del paisaje en el cantón de Hojancha, Costa Rica, Recur. Nat. Y. Ambient. CATIE (45) (2007) 81–90.
[33] IUCN - International Union for Conservation of Nature and Natural Resources. (2020). Global Standard for Nature-based Solution. Gland, Switzerland.
[34] E.J. Leguía, B. Locatelli, P. Imbach, C.J. Pérez, R. Vignola, Servicios ecosistémicos e hidroenergía en Costa Rica, Ecosistemas 17 (1) (2008).

[35] E. Lianes, M. Marchamalo, M. Roldán, Evaluación del factor C de la RUSLE para la gestión de la cobertura vegetal en el control de erosión en la cuenca del río Birrís, Costa Rica, Agron. ía Costarric. 2 (33) (2009) 217–235.
[36] Loría Chaves, A.M. (2023). Funcionalidad de bosques ribereños usando mamíferos silvestres como indicadores en el corredor biológico Montes del Aguacate. [Tesis de maestría, Universidad de Costa Rica]. https://hdl.handle.net/10669/9005.
[37] B. Luiza Amante, Conectividad funcional y servicios ecosistémicos provistos por abejas en paisajes agrícolas del Corredor Biológico Volcánico Central–Talamanca, [Tesis De. Maest. ía En. Pr. áctica De. Conserv. óN. De. la Biodivers. CATIE]. (2020). https://repositorio.catie.ac.cr/handle/11554/9856.
[38] Martínez-Rodríguez, M.R., Viguera, B., Donatti, C.I., Harvey, C.A., & Alpízar, F. (2017). Cómo enfrentar el cambio climático desde la agricultura: Prácticas de adaptación basada en ecosistemas (EbA), módulo 4. CATIE. https://repositorio.catie.ac.cr/handle/11554/9481.
[39] A. Mata Morales, C. Esquivel Rodríguez, M. Gutiérrez Miranda, F.E. Salas Hernández, Humedales resilientes: Una vía para promover la intervención pública a través de la gestión del riesgo en Costa Rica, Ambientico (296) (2019) 28–33.
[40] Mereles, L., Tranoy, C., Añino, I., Meza, L., & Ríos-Castillo, I. (2025). Transformación sostenible de sistemas alimentarios mediante soluciones basadas en la naturaleza: Experiencias y aprendizajes en la región del Sistema de la Integración Centroamericana. Panamá. FAO. https://doi.org/10.4060/cd4948es.
[41] Meza, L.E., Rodríguez, A.G. (2022). Soluciones basadas en la naturaleza y la bioeconomía: contribución a una transformación sostenible e inclusiva de la agricultura y a la recuperación pos-COVID-1. Serie Recursos Naturales y Desarrollo, (210), Santiago, Comisión Económica para América Latina y el Caribe (CEPAL).
[42] M. Miranda Quirós, M. Gutiérrez Miranda, E. Calderón Jiménez, K. Jiménez Elizondo, El Corredor Biológico Interurbano María Aguilar y el desarrollo de soluciones basadas en la naturaleza, Ambientico 281 (2022) 36–44.
[43] A. Mondragón-Rodríguez, A. Quesada-Román, Integrating geomorphological and socio-spatial factors for landslide risk zonation in Atenas, Costa Rica, Discov. Hazards 2 (1) (2026) 8.
[44] V.A. Neumann, J. Hack, A methodology of policy assessment at the municipal level: Costa Rica's readiness for nature-based stormwater management, Sustainability 12 (1) (2020) 230, https://doi.org/10.3390/su12010230.
[45] Okot, T. (2019). Análisis de iniciativas locales de prácticas agroconservacionistas para neutralizar la degradación de tierras en la cuenca del río Barranca, Costa Rica. Tesis de maestría, CATIE. https://repositorio.catie.ac.cr/handle/11554/9088.
[46] Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C. D., & Moher, D. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. bmj, 372.
[47] H.O. Pörtner, D.C. Roberts, E.S. Poloczanska, K. Mintenbeck, M. Tignor, A. Alegría, A. Okem, Summary for policymakers. *IPCC 2022: Climate Change 2022: Impacts*, Adapt. Vulnerability. Contrib. Work. Group. II Sixth Assess. Report. Intergov. Panel Clim. Change (2022) 3–33.
[48] M. Potthast, S. Geppert, Corredores biológicos interurbanos: Fusionando capital natural y construido, Ambientico (272) (2019) 5–13.
[49] A. Quesada-Román, Priorities for natural disaster risk reduction in Central America, PLOS Clim. 2 (3) (2023) e0000168.
[50] A. Quesada-Román, Mass Movements Dynamics and Morphologies in Costa Rica. Landscapes and Landforms of Costa Rica, Springer International Publishing, Cham, 2024, pp. 79–89.
[51] Adolfo Quesada-Román, Challenges and opportunities of global change for sustainability in Central America, Epub August 04, 2025, Rev. De. Cienc. Ambient. 60 (1) (2026) 21503, https://doi.org/10.15359/rca.60-1.2.
[52] A. Quesada-Román, J.A. Ballesteros-Cánovas, S. Guillet, J. Madrigal-González, M. Stoffel, Neotropical Hypericum irazuense shrubs reveal recent ENSO variability in Costa Rican páramo, Dendrochronologia 61 (2020) 125704.
[53] A. Quesada-Román, H.G. Hidalgo, E.J. Alfaro, Assessing the impact of tropical cyclones on economic sectors in Costa Rica, Central America, Trop. Cyclone Res. Rev. 13 (3) (2024) 196–207.
[54] A. Quesada-Román, E. Mata-Cambronero, The geomorphic landscape of the Barva volcano, Costa Rica, Phys. Geogr. 42 (3) (2021) 265–282.
[55] A. Quesada-Román, N. Montalván-Burbano, Global trends in scaling nature-based solutions for disaster risk reduction, Phys. Chem. Earth Parts A/B/C. (2026) 104302.
[56] A. Quesada-Román, N. Montalván-Burbano, M.E. Camacho-Umaña, N. Kariminejad, A. Islam, F. Domazetovic, Soil erosion in geomorphological studies: Recent trends and future directions, Progress. Phys. Geogr. Earth Environ. 49 (4) (2025) 443–463.
[57] A. Quesada-Román, D. Pérez-Umaña, A. Brenes-Maykall, Relationships between COVID-19 and disaster risk in Costa Rican municipalities, Nat. Hazards Res. 3 (2) (2023) 336–343.
[58] A. Quesada-Román, A. Picado-Monge, J. Rivera-Solís, M. Hernández, L. Torres-Berhard, M. Ruiz, Flood risk method for scarce-data catchments and municipalities, Rev. Geol. ógica De. Am. érica Cent. (2025) *72*.
[59] Ramírez Amador, C.J. (2020). Plan estratégico de restauración ecológica para el CATIE, Costa Rica. [Tesis de maestría, CATIE].
[60] Reid, P.H., de Madrid, M.P., & Ramírez, O. (2018). Ecosystem-based adaptation approaches: Strengthening evidence and informing policies. WCMC, UNEP, IIED & IUCN.
[61] C. Rodríguez-Arias, Cambios en estructura vertical y composición florística tras 15 años de restauración ecológica en la Reserva Madre Verde, Costa Rica, Rev. De. Cienc. Ambient. 57 (2) (2023).
[62] H. Rodriguez-Bolaños, K. Córdoba-Leitón, S. Zogaris, B.W. Mutaqin, N. Kariminejad, A. Islam, A. Quesada-Román, Land Use Change Alters Hydrological Drivers of Soil Erosion in the Tapesco River Watershed, Zarcero, Costa Rica, Hydrol. Process. 39 (12) (2025) e70355.
[63] Rodríguez-Bolaños, H., Córdoba-Leitón, K., Kariminejad, N., Mutaqin, B.W., Islam, A., Domazetović, F., … & Quesada-Román, A. (2026). Assessing soil erosion in volcanic landscapes using RUSLE approach: A long-term estimation (1986-2019) in the Reventado River watershed, Cartago, Costa Rica. Soil Security, 100234.
[64] J. Samaniego, S. Lorenzo, E.R. Toro, L.F. Krieger Merico, J. Herrera Jiménez, P. Rouse, N. Harrison, Nature-based solutions and carbon dioxide removal, CEPAL, 2022.
[65] L.F. Sandoval Murillo, C. Morera Beita, Corredores biológicos en Costa Rica: Metodología para monitoreo de estructura y conectividad del paisaje, Rev. Geogr. áfica De. Am. érica Cent. 75 (2025) 123–150.
[66] M. Schlönvoigt, La importancia de corredores biológicos para conservación de biodiversidad y adaptación al cambio climático, Ambientico (272) (2019) 13–19.
[67] Schueler, K. (2017). Soluciones basadas en la naturaleza para fortalecer la resiliencia costera. Banco Interamericano de Desarrollo. https://doi.org/10.18235/0000824.
[68] UNDRR - United Nations Office for Disaster Risk Reduction. (2025). *Global Assessment Report on Disaster Risk Reduction 2025: Resilience Pays: Financing and Investing for Our Future.* Stylus Publishing, LLC.
[69] N. Veas-Ayala, M. Alfaro-Córdoba, A. Quesada-Román, Costa Rican wetlands vulnerability index, Progress. Phys. Geogr. Earth Environ. 47 (4) (2023) 521–540.
[70] C.C. Vega, Soluciones basadas en la naturaleza dentro de políticas públicas ambientales, Ambientico (280) (2021) 9–14.